

**DESARROLLO DE TECNOLOGIAS Y SISTEMAS SpA**

**CERTIFICADO DE CALIBRACIÓN**

Laboratorio de Calibración

**N° de Certificado** : 25-MD-44-0136 **FECHA DE EMISION:** 28-mar-25 **Página 1 de 5**

**Descripción del Ítem**

**Cliente** : FUERZA AEREA DE CHILE IV BRIGADA AEREA, PUNTA ARENAS

**Unidad Interna** : GRUPO DE MANTENIMIENTO N° 53

**Descripción del ítem** : TEST SET ELECTRICAL POWER

**Fabricante** : NORTHROP

**Modelo / Número de Parte** : 76823-14-76024-1 **Número de Serie** : 0031

**Datos de la Calibración**

**Fecha de Calibración** : 25-mar-25 **Fecha de Vencimiento** : 25-mar-26

**Lugar de Calibración** : BANDADA ELECTRONICA

**Condiciones Ambientales** : **Temperatura** : 28,8 ° C **Humedad Relativa** : 24%

**Procedimiento** : 33A1-7-237-1

**Intervalo de calibración** : 12 Meses

**Antecedentes del o los Patrones Utilizados**

| Descripción         | Fabricante | Modelo   | N° de Serie | N° DTS   | Vence     | Emisor | Trazabilidad |
|---------------------|------------|----------|-------------|----------|-----------|--------|--------------|
| TRUE RMS MULTIMETER | FLUKE      | 87V      | 92010032    | 105951   | 26-ene-26 | DTS    | NIST         |
| TERMOHIGROMETRO     | VETO       | A6031000 | TEMP-003    | NO TIENE | 06-ene-26 | DTS    | NIST         |
| ---                 | ---        | ---      | ---         | ---      | ---       | ---    | ---          |
| ---                 | ---        | ---      | ---         | ---      | ---       | ---    | ---          |
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Los patrones utilizados en la calibración cuentan con trazabilidad a patrones nacionales y/o internacionales los que a su vez están referidos a patrones primarios de acuerdo al Sistema Internacional de Unidades (SI).

Los resultados de la calibración están relacionados con el ítem calibrado, referidos al momento y condiciones en las cuales fueron realizadas las mediciones.

Este Certificado de Calibración no puede ser reproducido total o parcialmente, excepto con el permiso del Laboratorio emisor.

El Laboratorio no asume responsabilidad por daños posteriores a la calibración, ocasionados por mal empleo o manipulación del instrumento.

Certificados sin firma ni sello de agua no son válidos.

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RESULTADOS DE LA CALIBRACIÓN

| FUNCION RANGO                    | COD. PATRÓN | VALOR ESTANDAR | LECTURA TMDE |       | ERROR | TOLERANCIA PERMITIDA ± |
|----------------------------------|-------------|----------------|--------------|-------|-------|------------------------|
|                                  |             |                | INICIAL      | FINAL |       |                        |
| OPERATIONAL CHECKOUT             |             |                |              |       |       |                        |
| FASE A 115V PINES B - D          |             |                | BRT          | DIM   |       |                        |
| FASE B 115V PINES M - F          |             | L-GEN          | ON           | ON    | N/A   |                        |
| FASE C 115V PINES K - H          |             | R-GEN          | ON           | ON    | N/A   |                        |
|                                  |             |                |              |       |       |                        |
| 115 V PINES FASE (D ) - NEUTRO R |             | V              | V            | V     | V     | V                      |
| R BUS A                          |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS B                          |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS C                          |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| R GEN                            |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| STATIC INV                       |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| L GEN                            |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS C                          |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 116          | 116   | 1     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS B                          |             | V              | V            | V     | V     | V                      |
|                                  |             | 115            | 115          | 115   | 0     | 3                      |
|                                  |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                                  |             | 400            | 400          | 400   | 0     | 4                      |

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**RESULTADOS DE LA CALIBRACIÓN**

| FUNCION RANGO             | COD. PATRÓN | VALOR ESTANDAR | LECTURA TMDE |       | ERROR | TOLERANCIA PERMITIDA ± |
|---------------------------|-------------|----------------|--------------|-------|-------|------------------------|
|                           |             |                | INICIAL      | FINAL |       |                        |
| OPERATIONAL CHECKOUT      |             |                |              |       |       |                        |
| L BUS A                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
|                           |             |                |              |       |       |                        |
| 115 V PINES B;M;K;D;F;H;G |             |                |              |       |       |                        |
| J;E.                      |             |                |              |       |       |                        |
| R GEN                     |             | V              | V            | V     | V     |                        |
| EXT TESTER                |             | 0              | 0            | 0     | 0     | N/A                    |
| VOLTMETER                 |             | 115            | 115          | 115   | 0     | N/A                    |
|                           |             | Hz             | Hz           | Hz    | Hz    |                        |
| FRECUENCIMETRO            |             | 400            | 400          | 400   | 0     | N/A                    |
| EXT METER C.B. ON         |             |                |              |       |       |                        |
| EXT TESTER                |             | V              | V            | V     | V     |                        |
|                           |             | 115            | 115          | 115   | 0     | N/A                    |
| 115 VAC CKT SEL           |             |                |              |       |       |                        |
| R BUS A                   |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS B                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS C                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| R GEN                     |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| STATIC INV                |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
|                           |             |                |              |       |       |                        |

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RESULTADOS DE LA CALIBRACIÓN

| FUNCION RANGO             | COD. PATRÓN | VALOR ESTANDAR | LECTURA TMDE |       | ERROR | TOLERANCIA PERMITIDA ± |
|---------------------------|-------------|----------------|--------------|-------|-------|------------------------|
|                           |             |                | INICIAL      | FINAL |       |                        |
| L BUS C                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS B                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS A                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| 115 V PINES B;M;K;D;F;H;G |             |                |              |       |       |                        |
| J;A.                      |             |                |              |       |       |                        |
| 115 VAC CKT SEL           |             | V              | V            | V     | V     | V                      |
| R BUS A                   |             | 115            | 115          | 115   | 0     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS B                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| R BUS C                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| STATIC INV                |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| L GEN                     |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS C                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |
|                           |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                           |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS B                   |             | V              | V            | V     | V     | V                      |
|                           |             | 115            | 116          | 116   | 1     | 3                      |

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**RESULTADOS DE LA CALIBRACIÓN**

| FUNCION RANGO          | COD. PATRÓN | VALOR ESTANDAR | LECTURA TMDE |       | ERROR | TOLERANCIA PERMITIDA ± |
|------------------------|-------------|----------------|--------------|-------|-------|------------------------|
|                        |             |                | INICIAL      | FINAL |       |                        |
|                        |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                        |             | 400            | 400          | 400   | 0     | 4                      |
| L BUS A                |             | V              | V            | V     | V     | V                      |
|                        |             | 115            | 117          | 117   | 2     | 3                      |
|                        |             | Hz             | Hz           | Hz    | Hz    | Hz                     |
|                        |             | 400            | 400          | 400   | 0     | 4                      |
| 28 V J1 PINES P+ ; N - |             | V              | V            | V     | V     | V                      |
| R BUS A                |             | 28             | 28           | 28    | 0     | 1                      |
| R BUS B                |             | 28             | 28           | 28    | 0     | 1                      |
| R BUS C                |             | 28             | 28           | 28    | 0     | 1                      |
| R GEN                  |             | 28             | 28           | 28    | 0     | 1                      |
| STATIC INV             |             | 28             | 28           | 28    | 0     | 1                      |
| L GEN                  |             | 28             | 28           | 28    | 0     | 1                      |
| L BUS C                |             | 28             | 28           | 28    | 0     | 1                      |
| L BUS B                |             | 28             | 28           | 28    | 0     | 1                      |
| L BUS A                |             | 28             | 28           | 28    | 0     | 1                      |




MARCO DÍAZ DELGADO  
TÉCNICO METRÓLOGO

Fin del certificado de calibración